

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2019

ESE 3211

(Coal Power Generation)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is coal? How coal is formed? Briefly explain. 07
- 1(b). What are the available sources of coal in Bangladesh? Briefly describe the present electricity generation situation in Bangladesh by using coal. 13
- 1(c). A coal sample has the following results: 15
- (i) 0.5 g coal produces 1.35 g of CO₂ and 0.225 g of H₂O.
- (ii) 0.4 g coal liberates ammonia which was absorbed in 50 mL of 0.5N H₂SO₄. The resultant solution required 22 mL of 1N NaOH solution for complete neutralization.
- (iii) combustion of 0.5 g coal in bomb calorimeter by the treatment with BaCl₂ produces 0.06 g of BaSO₄.
- Calculate the percentage of C, H, N, S and O by assuming ash content in the coal is 4.5%. Also find HCV and LCV.
- 2(a). Write short notes on: (i) out-plant coal handling, and (ii) in-plant coal handling. 08
- 2(b). Why coal cleaning is necessary? With a schematic diagram, give a brief description of rotary drying process of coal? 10
- 2(c). What is coal preparation? What are the basic types of operation used for cleaning the coal? 07
- 2(d). What are bottom ash and fly ash? Write down the significant applications of bottom ash and fly ash. 10
- 3(a). With a schematic diagram, describe the hydraulic system for bottom ash handling. 08
- 3(b). What are the different types of dust collector? Describe the electrostatic precipitator system of dust collection. 08
- 3(c). What are the factors must be considered for selecting a dust collector? 04
- 3(d). Give a brief description of different coal firing methods. Which one is preferred for high volatile coal and why? 15
- 4(a). Give a brief description of fixed bed coal combustion technology. 10
- 4(b). Why coal is pulverised before combustion? Describe the ring and ball mill coal pulveriser with neat sketch. 12
- 4(c). What are the essential requirements of pulverised coal burners? 05
- 4(d). Write down the types of pulverised coal burner. Draw the front and side view of cyclone burner. 08

SECTION – B

- 5(a). What is clean coal technology? With a flow diagram, show the processes of a typical coal cleaning plant. 10
- 5(b). What are the various types of technology used to capture CO₂? Describe the working principle of Oxy-fuel combustion system for CO₂ capture. 10
- 5(c). With the help of a flow diagram, describe the integrated gasification combined cycle (IGCC) plant with CO₂ capture. 10
- 5(d). What are the methods currently used for CO₂ capture in post-combustion process? 05
- 6(a). Why CO₂ is compressed before transportation it to the storage location? With neat sketch, describe the geological sequestration of CO₂. 12
- 6(b). Describe the wet scrubber process to capture the SO₂ from flue gases. 10
- 6(c). What are the various technologies that have been developed for primary control of NO_x? Describe the selective catalytic reduction method to capture NO_x from flue gases. 13
- 7(a). What does mean by steam generator? Compare between water tube and fire tube boiler. 07
- 7(b). Why boiler mountings and accessories are used in boiler? 05
- 7(c). Describe the working principle of La-Mont boiler with the help of schematic diagram. 13
- 7(d). What is equivalent evaporation? 10
- In a steam power plant, quantity of steam generation is 700 kg/h, quantity of fuel used is 534 kg/h, gross calorific value of the fuel is 40375 kJ/kg, the enthalpy of steam and feed water are 2782 kJ/kg and 176 kJ/kg, respectively. Calculate the equivalent evaporation and boiler efficiency.
- 8(a). Why nozzles are used in steam turbine? Classify them with sketch. 05
- 8(b). Why compounding of turbine is done? For a multistage steam turbine, with the help of h-s diagram, show the condition line and define the reheat factor. 08
- 8(c). What is meant by steam turbine governing? With necessary diagram, briefly explain the throttle governing process of steam turbine. 10
- 8(d). A Parson's turbine consists of one row of fixed blades and one row of moving blades. 12
- The blade speed is 110 m/sec, mass of steam passing per second is 13.5 kg, steam velocity at exit from fixed blades is 144 m/sec, blade outlet angle is 20°. Assume the efficiency of conversion of heat energy into kinetic energy in the moving blade channels as 0.92. Calculate the (i) power developed in the stage, (ii) stage efficiency and (iii) maximum efficiency.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2019

ESE 3207

(Solar Photovoltaic Systems)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | Why is the Fermi level of an n-type semiconductor near to the lowest conduction band energy, and for a p-type semiconductor near to the highest valence band energy? What is direct and indirect bandgap semiconductor? | 05 |
| 1(b). | What are the four main ideas represented in Quantum theory? Explain surface recombination, and mention its significance in a solar cell. | 10 |
| 1(c). | Discuss about the different types of losses in a solar cell. | 10 |
| 1(d). | Explain the metal-semiconductor junction with energy band diagram under different bias conditions. | 10 |
| 2(a). | Explain the Auger recombination in detail. | 10 |
| 2(b). | Draw and discuss the equivalent circuit of an ideal solar cell and a actual solar cell with series resistance R_s and shunt resistance R_p . | 10 |
| 2(c). | Discuss in details about the balance limit of solar cell efficiency according to Schokley-Queisser limit. | 15 |
| 3(a). | Discuss the production of silicon wafer from quartzite. | 14 |
| 3(b). | Discuss the fabrication of C-Si solar cell from silicon wafer. | 14 |
| 3(c). | What is TCO? Explain. | 07 |
| 4(a). | Mention the names of some III-V materials used in thin-film solar cell production. | 03 |
| 4(b). | What is lattice mismatch? Justify the use of Ge, GaAs, and GaInP to form triple-junction cells. | 07 |
| 4(c). | Why hydrogen is used in thin-film silicon alloys? Why thin-film silicon solar cell is called a drift device? | 10 |
| 4(d). | Discuss about the design of thin-film silicon solar cells. | 15 |

SECTION – B

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|-------|---|----|
| 5(a). | Write short notes on (i) PV module, and (ii) PV array. | 04 |
| 5(b). | Explain the working principle of photovoltaic panel in series and parallel connection system. Why series and parallel connections are necessary in PV system? | 12 |
| 5(c). | Explain the basic working principle of a PV solar battery charging technique using PWM method. | 10 |
| 5(d). | Explain the working principle of a PV powered water pumping system. | 09 |

6(a).	Write short notes on (i) specific gravity, (ii) autonomy days, (iii) luminous efficiency, and (iv) total harmonic distortion (THD).	08
6(b).	Write the merits and demerits of PV pumping system when it is directly connected to PV array or to a battery.	06
6(c).	Discuss the different types of cost of PV power generation system.	06
6(d).	The sample of the load profile from the SHS using a customer is i) laptop computer 2 numbers, each 20 watt, daily operation 3 hours, ii) electric lamp 4 numbers, each 30 watt, daily operation 8 hours, iii) colour television 2 numbers, each 50 watt, daily operation 6 hours, iv) gizer 1 number of 800 watt, daily operation 2 hours. The system inverter power factor is unity, DOD = 60%, autonomy days is 4, battery charging efficiency is 75%, peak sun hours daily is 5, battery voltage is 12 V. Using the above mentioned data, design a PV module. Also, calculate (i) battery capacity, (ii) charge controller capacity, (iii) DC to AC inverter capacity, (iv) switch capacity, and (v) wire size.	15
7(a).	Discuss the working principle of on-grid and off-grid solar system with necessary diagram.	10
7(b).	What is Net Metering? How does it work?	11
7(c).	What are the conditions for grid interfacing of on-grid PV system?	06
7(d).	Write down the eligibility conditions for a customer to be considered for Net Metering connection in Bangladesh.	08
8(a).	Discuss the effects of PV module temperature, orientation and tilt angle on the performance of a solar PV systems?	10
8(b).	Why a charge controller is important in a solar PV system? Write the operation principle of shunt regulator and series regulator.	10
8(c).	What are the basic parameters of a charge controller?	05
8(d).	Discuss about the testing, installation, and replacing procedures of a solar home system with different components.	10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2019

ESE 3203

(Petroleum and Natural Gas Processing)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Briefly describe crude oil compositions. What are the impurities that are associated with a crude oil? Give examples of each of them. 15
- 1(b). Classify petroleum products based on boiling point and carbon numbers. 08
- 1(c). What is octane number? Explain the effect of variables on an engine's octane rating. 07
- 1(d). Classify diesel fuel in terms of engine speed. 05
- 2(a). What is catalytic reforming? Briefly describe the major reactions in a catalytic reforming process. 10
- 2(b). Draw the process diagram of the SR reforming process. Explain the variation of temperature and effluent compositions in the reactors of the SR process. 12
- 2(c). Distinguish between CCR and SR reforming processes. 08
- 2(d). Write down C₅ and C₆ isomers and mention their RON values. 05
- 3(a). Describe following terms in brief. 09
(i) Total acid number, (ii) TBP distillation, and (iii) Natural gas condensate
- 3(b). What are the objectives of product blending? Outline how to calculate RVP blending in the product. 08
- 3(c). How one can estimate the viscosity of a product blend? Estimate the viscosity of a blend of 2000 barrels of Fraction-1 with a viscosity of 75 cSt, 3000 barrels of Fraction-2 with a viscosity of 100 cSt, and 4000 barrels of Fraction-3 with a viscosity of 200 cSt, all at 130 °F. 18
- 4(a). What are the objectives of light naphtha isomerization? 05
- 4(b). Write down the main and side reactions of an isomerization process. 10
- 4(c). Light naphtha with a specific gravity of 0.724 is used as a feed to the isomerization unit with a feed rate of 1000 m³/h. The yields of converted components are given in the following table. Find the product composition considering both converted and unconverted products. 08

Component	Yield (wt %)
C ₃	0.348
iC ₄	0.619
nC ₄	1.770
C ₅ ⁺	97.261

- 4(d). With necessary diagram, describe crude oil desalting process. What are the operating variables of desalter? 12

SECTION – B

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|-------|--|----|
| 5(a). | What is LNG? Describe its properties. With a schematic of a LNG plant, describe the overall LNG production process. | 20 |
| 5(b). | Write short description of different types of LNG plants. | 06 |
| 5(c). | Discuss in details, why at least three refrigerants are necessary to liquefy natural gas. | 09 |
| 6(a). | With proper process schematic and T-s diagrams of refrigerants, discuss cascade type LNG production process in details. | 20 |
| 6(b). | Write short notes on the following: | 15 |
| | (i) Gas quality aspect of LNG | |
| | (ii) Wobbe index | |
| | (iii) Associated and non-associated gas | |
| 7(a). | Write a description about kettle type shell and tube heat exchanger with simplified schematic diagram. | 13 |
| 7(b). | Draw the phase envelop of gas components and hydrocarbon mixture. In that diagram denote bubble point line, dew point line, critical point, cricondenbar, cricondentherm, and retrograde area. | 12 |
| 7(c). | What does flow assurance mean in terms of natural gas transport? What is natural gas hydrates and why they form? | 10 |
| 8(a). | Draw a schematic of a three phase separator. Describe in details, the operation of this unit in petroleum production. | 15 |
| 8(b). | Describe natural gas dehydration process by adsorption process with necessary schematic diagram. | 10 |
| 8(c). | Describe in details, the LPG and NGL extraction process with necessary process flow diagram. | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2019

ESE 3221

(Energy Storage Systems)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What are the needs of energy storage? Mention different modes of energy storage. 04
- 1(b). Write short notes on: (i) Energy density, (ii) Power density, and (iii) Specific power. 06
- 1(c). What is pumped storage power plant? Showing schematics of a pumped storage facility, explain its working principle. 12
- 1(d). What is flywheel energy storage system? What are the advantages and disadvantages of flywheel energy storage system? How would you compare flywheel energy storage system with other mechanical electricity storage technologies? 13
- 2(a). Illustrate with details, a aquifer thermal energy storage system. 10
- 2(b). How does compressed air energy storage work? Explain with diagram. 08
- 2(c). Classify and explain about different types of superconductor. 06
- 2(d). Discuss about the operating principle, application, and common challenges of SMES. 11
- 3(a). Show the comparison between battery and capacitor. What are the differences between capacitor and superconductor? 10
- 3(b). Derive the equation of capacitance for an EDLC. 15
- 3(c). Derive the equation of magnetic field of a toroid. Why SMES is made in toroidal shape? 10
- 4(a). Summarize the state of most energy storage technologies based on their primary applications, what we know currently about them, and what are the challenges. 10
- 4(b). Discuss about different aspects of storage system as grid component. 15
- 4(c). What are the key barriers of widespread deployment of energy storage as a grid component? 10

SECTION – B

- 5(a). What are the differences between cell and battery? Discuss the working principle of lead-acid battery. 10
- 5(b). Briefly explain the working principle of sodium-sulphur battery and mention the current challenges associated with this battery. 10
- 5(c). “We should provide higher voltage than the battery voltage to charge the battery” – why? Explain. 08
- 5(d). Write down the advantages and disadvantages of lead-acid battery. 07

6(a).	What is solid electrolyte interphase (SEI) layer? Mention briefly the root cause of degradation and various degradation modes in a battery.	10
6(b).	How does lithium-ion battery work – discuss using thematic diagram, along with relevant anodic and cathodic reactions?	10
6(c).	Briefly discuss the working principle of vanadium redox flow battery. Can vanadium flow battery beat lithium-ion battery in high scale uses? Explain.	15
7(a).	Classify fuel cells based on electrolytes. Draw a thematic diagram of proton-exchange membrane fuel cell, along with anodic and cathodic reactions.	10
7(b).	Briefly explain, how you would store hydrogen fuel in a small fuel cell system.	08
7(c).	Name the common fuels or chemicals that considered a candidate fuel in small fuel cells.	07
7(d).	What are the common applications of proton-exchange membrane fuel cells? Write down the advantages and disadvantages of proton-exchange membrane fuel cells.	10
8(a).	Draw a schematic diagram on how to connect multiple fuel cells in a series to obtain practical voltage level.	10
8(b).	Illustrate the alkaline fuel cell with reactions. What are the differences between alkaline and acid fuel cells?	10
8(c).	Write short notes on:	08
	(i) Thermal management in fuel cells	
	(ii) Lithium-polymer battery	
8(d).	Briefly mention the system requirements for a small fuel cell.	07

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2019

ESE 3217

(Instrumentation and Control)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

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iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). With necessary block diagram and time frame associated with each block, write a description about hierarchy of process control activities. 20
- 1(b). Discuss the four different ways that a dynamic model is used in process control engineering. 10
- 1(c). Write short note on modern differential pressure cells. 05

- 2(a). Develop a dynamic model of a stirred tank heating process with constant hold up. 10
- 2(b). Define Laplace transform. Show that this transformation is linear. 10
- 2(c). Solve the following differential equation using Laplace transform. 15

$$5 \frac{dy}{dt} + 4y = 2 \quad \text{when, } y(0) = 1.$$

- 3(a). Using inverse Laplace transform, find $y(t)$ for 10

$$Y(s) = \frac{s+1}{s^2(s^2+4s+5)}$$

- 3(b). Find the transfer function for the following ODE: 15

$$\frac{d^2y}{dt^2} + 6 \frac{dy}{dt} + 9y = \cos(t) \quad \text{for } y(0) = 1 \text{ and } y'(0) = 2$$

- 3(c). Linearize the following ODE: 10

$$\frac{dy}{dt} = x^3 + 1 + \sqrt{u} \quad \text{at } x = 2, \quad u = 5$$

- 4(a). An electrically heated, stirred tank model is given by following system of ODEs. 10

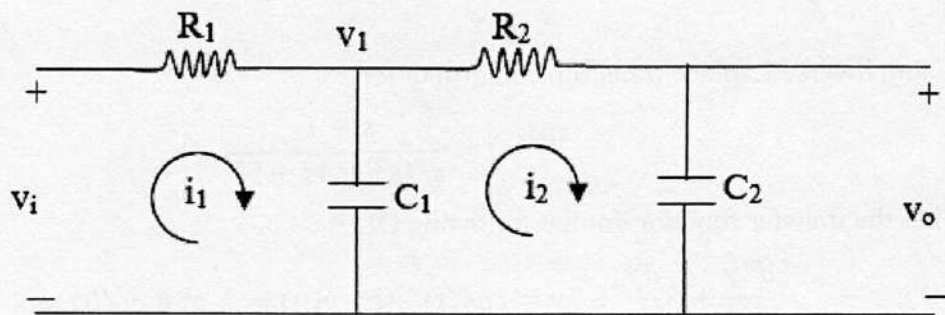
$$\begin{bmatrix} \dot{T}' \\ \dot{T}_e' \end{bmatrix} = \begin{bmatrix} -0.2 & 0.1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} T' \\ T_e' \end{bmatrix} + \begin{bmatrix} 0 \\ 0.05 \end{bmatrix} Q'$$

Show that the system is stable.

- 4(b). Draw a schematic of a feedback control system and describe its basic operating principle. 05
- 4(c). What are the basic category of sensors used in process and energy industry? Write brief description. 08
- 4(d). What is a PID controller? Describe the PID controller operation with corresponding equations and necessary graphs. 12

SECTION – B

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|-------|--|----|
| 5(a). | Write short note on microcontroller. Draw the block diagram of a microcontroller structure and discuss the working principle of a microcontroller. | 15 |
| 5(b). | Define SCADA system. Explain the working procedure of a SCADA system. | 08 |
| 5(c). | What is meant by Distributed Controller System (DCS)? Describe the elements of DCS. | 08 |
| 5(d). | Define: (i) Instrumentation and (ii) Control engineering. | 04 |
| | | |
| 6(a). | Draw the complete block diagram of a PLC system with main functional items. Also, explain the functions of each block. | 14 |
| 6(b). | What is a stepper motor? Explain the working principle of a stepper motor. | 12 |
| 6(c). | Explain different types of PLC programming. | 09 |
| | | |
| 7(a). | Draw and write the equivalent logic diagram of AND gate, OR gate and NOT gate using PLC ladder logic programming language. | 12 |
| 7(b). | What is an internal relay? Why is it important in a PLC system? | 05 |
| 7(c). | Why timers are essential in a PLC system? Explain the different types of timers. | 08 |
| 7(d). | What are analog and digital signals? Draw the block diagram of a communication process. | 10 |
| | | |
| 8(a). | What is a feedback system? Derive the necessary equations and draw a block diagram for the two-section R-C circuit of the following figure. | 10 |



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|-------|--|----|
| 8(b). | According to the number of inputs, classify the open-loop configuration of OP-amp. Also, draw and explain each of the configuration. | 10 |
| 8(c). | What is fuzzy logic? Describe the basic working principle of fuzzy logic with block diagram. | 10 |
| 8(d). | What is modulation? Why modulation is used in instrumentation and control? | 05 |

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